

PRASHANT KUMAR

+91 8477993577 | prashantkumar_24ci01@dtu.ac.in | extbar
github.com/prashantkumar24ci01-bliplinkedin.com/in/prashant-kumar-3551b6214 extbar github.com/prashantkumar24ci01-bliplinkedin.com/in/prashant-kumar-3551b6214
M.Tech (Control & Instrumentation) — Delhi Technological University | GATE 2024 Qualified (EE)
AI Catalyst Intern — Havells CRI

EDUCATION

Degree	Year	Institution	CGPA/%
M.Tech (C&I)	2024–26	Delhi Technological University	74.5%
B.Tech (EE)	2017–21	MJP Rohilkhand University, Bareilly	69.2%

TECHNICAL SKILLS

AI & ML Stack	RAG Pipelines, ChromaDB (Vector DB), Sentence Transformers, LLM Orchestration, NLP (Embeddings, Semantic Search), OpenRouter API, Model Evaluation
Programming	Python (NumPy, Pandas, Matplotlib, HuggingFace), MATLAB, Shell/Bash, Git
Engineering Tools	SPICE Simulation (ngspice, PSpice), Cadence Virtuoso, OrCAD, MATLAB/Simulink, Testbench Automation
Core Competencies	Workflow Analysis & Automation, Engineering Productivity, Design-of-Experiments, Signal Processing, Control Systems, Circuit Analysis
Platforms	Linux (Redhat, Debian, NixOS), Windows, WSL

PROJECTS

AnalogCheck — AI-Powered SPICE Error Checker with RAG Pipeline Capstone Project

Python, ChromaDB, Sentence Transformers, OpenRouter LLM, ngspice

- Built CLI tool that semantically checks SPICE netlists for common IC design errors, combining 5 deterministic health checks with an AI reasoning layer — directly analogous to engineering copilot deployment.
- Implemented RAG pipeline: 20 annotated IC error patterns + PSpice reference documents embedded via `all-MiniLM-L6-v2` into ChromaDB vector store.
- Integrated LLM reasoning (OpenRouter API) for error explanation and fix suggestion generation — demonstrates LLM orchestration, prompt engineering, and AI tool calling.
- Automated ngspice subprocess validation: parses simulation output, detects DC convergence failures, floating nodes, and missing ground — AI-driven engineering workflow automation.
- Generated structured JSON/Markdown reports; designed for CI/CD integration into design verification flows.

Engineering Workflow Automation: Bias-Sweep Testbench Generator

Python, ngspice, MATLAB

- Developed automated AC analysis testbench generator that sweeps 4 transistor bias conditions and extracts gain, phase margin, and -3 dB bandwidth — replaced manual per-condition testbench editing.
- Demonstrated $5\times$ reduction in simulation setup time; generated comparative performance reports programmatically.

M.Tech Thesis: Transistor-Level Analog Filter Verification

ngspice, Python, MATLAB, $0.18\ \mu\text{m}$ CMOS BSIM3v3

- Verified single-OTRA biquad filter at three abstraction tiers (symbolic, behavioral, transistor-level) — systematic multi-level validation methodology transferable to AI model evaluation frameworks.
- Identified and resolved high-pass passband droop via systematic DOE bias sweep across 4 operating points — data-driven root-cause analysis.

ADDITIONAL EXPERIENCE

Teaching Assistant — Dept. of Electrical Engineering, DTU Aug 2024 – Present

- Conducted VLSI and digital logic labs for 30+ students; developed automated testbenches for functional verification.

ACHIEVEMENTS

- GATE 2024 Qualified (Electrical Engineering) — top 15% nationally.
- Published IEEE Conference Paper: “Application & Performance Comparison of Electrical Motors in HEVs.”
- Built and deployed 2 AI/ML tools (AnalogCheck RAG pipeline + automated testbench generator) demonstrating end-to-end AI project execution — from problem identification to production-ready output.